

DSC 323: DATA ANALYSIS & STATISTICAL SOFTWARE II / DSC 423: DATA ANALYSIS AND REGRESSION

Final Project | Total Points 50

The purpose of the final project is to demonstrate your ability to apply the knowledge and the techniques learned during this course. Final project is done individually.

FINAL PROJECT:

I. Pick a Dataset (Lecture – 6)

[2 Points]

UNDERGRADUATE STUDENTS

You need to pick one of the datasets provided on D2L to complete your final project. These are very small, less complex datasets. Once you identified the dataset, familiarize yourself with the dataset. Review the description, and identify the dependent variable and independent variables. **The Dependent variable (DV) specified in the dataset should be used as your DV for the model. If you use a different variable, you will get a zero for the whole project. The main objective is for you to know which variable is used for which regression technique which we learnt from week-2 onwards. Also, you must use a dataset provided. Points will be deducted if you do not use the whole dataset or do not use all the variables in the dataset (except for ID columns). If you do not use a dataset provided, you will receive a zero.**

GRADUATE STUDENTS

Graduate students will have to find your own dataset. No two graduate students can use the same dataset. Therefore, once you finalize the dataset, post on D2L Discussion board under 'Dataset Selected'. Include your full name (as it appears on campusconnect), dataset name and the link to the dataset. Before you post the dataset, you have to check the discussion board to see if someone else has already taken the dataset. If its already taken, then you have to find another one. Best option would be to identify a few datasets to save time. Once you pick the dataset, you will not be able to change it. **None of the students can use Boston Housing Price dataset, datasets posted for undergraduate students even if it is available on other sites and also datasets used in class.** **Should have more than 500 observations without missing values. Minimum number of variables should be 6 or more excluding the DV variable and have at least one categorical variable which is not the DV. If you do not use all the variables and all the observations in the dataset, points will be deducted. If the dataset has more than 40K observations, you will be required to randomly select 10-20K observations.**

Proposal: Both undergraduate and graduate students

What to submit: Submit a single page document with the following information:

- (1) Dataset name
- (2) Link to the dataset (for undergraduates state 'D2L')
- (3) Screen capture of dataset with 5 observations. If the dataset has a lot of variables, you can provide as multiple parts.
- (4) What is the DV?
- (5) What are the IV? How many are there? List the IV names.
- (6) How many observations does the dataset have? Are there many missing values? If yes, include the number of observations and a screen capture.
- (7) Time line for project: E.g. list the each week from week-6 onwards, and specify what will be covered each week.

II. Review Dataset and Code (Lecture – 7 to 9)

Each week code at least 15-20 lines. This will help you stay on track and complete the project on time.

III. Coding, Analysis (Due Lecture - 10)

[20 Points]

This part includes the methodology and analysis (critical thinking) including the coding done to generate the necessary outputs to write your report. The code is evaluated based on the implementation of the right techniques in the right sequence. Make sure to review the instructions for the report's "Analysis, Results and Findings" section to make sure your code includes all the necessary steps. It is important to use the right technique for the dataset you selected. **Use of incorrect technique or implementing techniques not covered in class will result in significant amount of points deduction. Code should be submitted in .sas file to be able to run the code.**

IV. Presentation Slides and Presentation (Due Lecture – 10)

[10 Points]

By now you should have completed the coding and generated the required outputs.

UNDERGRADUATE STUDENTS

Create the presentation slides and upload it to D2L along with the SAS code and datafile. **You will not be required to present.**

GRADUATE STUDENTS

Create the presentation slides and upload it to D2L along with the SAS code and data file. Asynchronous students will provide a recording of the presentation while the in-person students will present in class during week-10.

Presentation Slides – All Students:

The main objective of the presentation to explain your research in a concise manner. Therefore:

- Presentation slides should include (1) introduction (2) dataset, variables etc., and (3) and important outputs to help explain your project and model including key findings listed in the report section of this document.
- Include the important outputs, and few bullet points describing the key findings of the output. If there are no bullet points describing the key findings on each slide, you will not receive points.
- Do not include any code, you will lose points.

Presentation and Recordings:

1. In-person Students:

- a. Should present in-person. Recorded presentations will not be accepted.
- b. Presentation should be 7 minutes long.

2. Asynchronous Students:

- a. Recordings should not exceed 7 minutes.
- b. The recording should be uploaded to D2L. Do not provide links to your Shared Drive, YouTube, OneDrive etc., it will result in zero points.**

V. Report (Due Lecture - 11)

[18 Points]

Note: Page length does not matter, if you need more pages to explain what you need to it is OK.

A Technical Summary Report: Report should be in an essay format and be about 5-10 pages. The appendix and SAS code do not count towards the page limit. SAS code should be submitted in a SAS file. It should also include all the important outputs in the appendix section. This report is intended for a statistically literate audience and must be written in a clear organized fashion using the correct terminology. The format should be like a report or journal article, as **NOT** like your assignment. It should consist of the following sections:

Introduction	Describe the goal or objective and any hypothesis, any literature review or background research you did using the references, why it is important, context, motivation etc.	2 Points
Analysis, Results and Findings	Your analysis should address the following points: Note: This is a list of techniques is provided only for guidance, and is not given in the order of execution. Students should review all the course materials to come up with a list of all techniques learnt in class, and implement the relevant techniques for your dataset in the right sequence. 1. The exploratory analysis of the data including descriptives that may suggest a possible model that is adequate for fitting the data. Do the data show a non-linear relationship? Should a transformation of the response variable and/or the predictors be useful? 2. Try interaction variables. 3. Use either linear, logistic, polynomial regression techniques and/or transformations 4. Check for collinearity among the independent variables. 5. A variable selection method will enable you to select suitable models and find the set of predictor variables, which are more informative for predicting the response variable. 6. You may want to fit a few models that seem adequate for your data and then select the model among them that provides the "best" prediction of Y. 7. Analyze the residual plots to look for patterns that might suggest a failure in the assumptions and some inadequacies in the selected model. 8. The existence of outliers and influential points may have dramatic effects on your analysis. Check also if there are outliers. 9. Can your model be improved? Are you satisfied with the model you have chosen? 10. Use the selected regression model to examine the relationship and associations among the variables in your study and to identify, among the observed independent variables, the strongest predictors for the response variable. 11. Compute two predictions including the prediction intervals using the regression model. 12. Apply validation techniques to evaluate the predictive power of your model. Split the original dataset at random into a training and test set. Test set should have at least 15 observations in order to compute meaningful validation statistics. Discuss the model performance using training, and testing sets. Hints for the Statistical Analysis: It is possible that you may not find a satisfactory model that fits adequately your data. Sometimes a data set may admit more than one satisfactory answer; sometimes there may be none. If the statistical analysis shows that no regression models are suitable for your data set, mention what approaches you have tried and what was unsatisfactory about them. If there is more than one suitable model, mention the pros and cons and compare their performance in predicting the response variable. The final aim of any statistical analysis is the understanding of a phenomenon or the investigation of a scientific problem, which your data arise from. Remember that the	16 Points

	regression function is a mathematical representation of such a problem and the interpretation of the parameters values will give you insights about the relationships of the variables in the problem.	
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IMPORTANT

- **When you submit any deliverable:**
 - Triple check your submission to see if you have submitted the right files.
 - Are they submitted in the correct format? E.g. .sas file, word file, .mp4 etc.
 - After you submit any deliverable, click on the submissions, it will open the files so you can check to see if they are the correct version.
 - **Project deliverables will not be regraded**, so make sure you are submitting the right versions / formats and also in the right submission folder.
- **Backups:**
 - Keep multiple backups of your code and the report each time you work on so that work will not be lost no matter what happens.
- **Code:**
 - If the code is not included you will receive a 0% as the code determines the analysis, and the outputs necessary for the report.
 - Your code should run for the dataset provided on D2L.
 - If the code does not execute, you will lose 50% of your grade. However, if the errors are at the beginning or at the time of importing, you will receive a 0% as code determines everything else.
 - If model testing is not included, 20% of the grade will be lost for each deliverable (code, presentation, report); the whole purpose of building the model is to see how well it predicts on unseen data and model testing help us do that.
- Do not provide zipped file, .tar or other compressed formats, if I am unable to open the files, you will receive zero.
- No late project will be accepted even if it is a couple of minutes late especially since you were allocated time to start working on the project from week-6.